

# SpatialLens Assist: Motion-Aware Hazard Detection for Campus Navigation from Low-FPS Egocentric Video

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## Abstract

An object detector tells a blind pedestrian *what* is nearby, but not whether it is closing, crossing, or simply present. We ask whether a small, interpretable set of motion cues—bounding-box growth, centroid displacement, optical flow, and ego-motion compensation—can recover useful hazard labels from low-fps handheld campus video, without training a learned trajectory model. We present **SpatialLens Assist**, an end-to-end CPU pipeline that converts short ( $\sim 10$ s) walkway clips into per-track motion-aware alerts over six classes (**approaching**, **crossing\_LTR**, **crossingRTL**, **moving\_away**, **static**, **uncertain**). The system pairs YOLOv8n with an IoU+centroid tracker, ego-motion-compensated Farneback flow and frame differencing, per-track motion aggregates, and a transparent rule cascade that emits a natural-language evidence string and treats **uncertain** as a first-class abstention rather than a guess. On 11 controlled walkway videos (61 manually labelled tracks), the system reaches **82.5%** overall accuracy on 57 evaluated track-label pairs and **93.3%** on the 30 tracks with at least three frames, with approaching precision/recall **0.95/0.88**—the safety-asymmetric regime an alert layer needs. Ablations show the largest gain comes from tracker continuity (+6.2pts from appearance-gated re-identification, +3.5pts each from two short-track salvage rules), and ByteTrack is a negative result on this sparse 2 fps regime (32.8%). The dominant failure mode is conservative abstention from tracker fragmentation, not dangerous mislabelling: *no approaching*  $\leftrightarrow$  *moving\_away* confusion occurs.

## 1 Introduction

The motivation for this work is personal. A few months ago my own glasses broke, and even the short experience of crossing campus with badly degraded vision made it strikingly hard to judge whether an approaching cyclist or pedestrian was on a collision course with me. For blind and low-vision pedestrians that difficulty is not a temporary inconvenience but a constant, and far more severe, daily reality. That gap is what motivated SpatialLens.

A bicycle parked against a wall and a bicycle bearing down on the camera are both detected as “bicycle, 0.86.” For a sighted user the wider visual context carries the intent; for a blind or low-vision pedestrian relying on an assistive device such as Seeing AI [7] or a wearable navigation aid [8], that context is exactly what is missing. The safety-critical signal is not the class label but the *motion*: whether the agent is closing, crossing the path, or merely present.

We frame motion-aware hazard detection as *per-track classification*: given a  $\sim 10$ s phone clip sampled at 2 fps, decide for each tracked object whether it is approaching, crossing left-to-right, crossing right-to-left, moving away, static, or explicitly **uncertain**. Three properties drive the design: (i) **explainability** – every prediction carries a textual evidence string; (ii) **calibrated abstention** – short tracks do not bluff, so **uncertain** is a first-class output with a sub-reason; (iii) **reproducibility on CPU** – the full pipeline runs in  $\sim 4$  min on a GPU-less Mac and the 61-track labelled set is committed.

**Summary.** On 11 controlled walkway videos and 61 labelled tracks the system reaches 82.5% accuracy on 57 matched tracks (93.3% on the 30 decidable tracks with  $\geq 3$  frames); **approaching** reaches 0.95 / 0.88 precision/recall and no **approaching**  $\leftrightarrow$  **moving\_away** confusion is observed. Isolated ablations attribute the dominant gain to track continuity rather than classifier complexity: appearance-gated

re-identification contributes 6.2pts and the two 2-frame salvage rules 3.5pts each, while ByteTrack on 2 fps offline footage drops to 32.8% – a useful negative result.

**Contributions.** (C1) a CPU-only offline egocentric pipeline for motion-aware campus hazard alerts; (C2) ego-motion-compensated motion features (median-subtracted Farneback flow + ECC-aligned frame differencing) appropriate to handheld low-fps footage; (C3) an interpretable hazard rule cascade with natural-language evidence strings, conservative abstention, and a trajectory-reversal heuristic for “approached-then-turned-away” tracks; (C4) a 61-track evaluation set with two reporting denominators and a fully scripted ablation suite; (C5) empirical evidence that tracking continuity dominates classifier sophistication in this regime, and that ByteTrack is poorly matched to sparse offline footage.

## 2 Related Work

**Detection and tracking.** YOLOv8 [1] is a fixed COCO-class front-end. SORT [3] and ByteTrack [2] are strong on-line baselines but assume near-continuous per-frame detections from a Kalman model; on our 2 fps footage ByteTrack (via **supervision** [4]) drops 51% of detections, motivating a transparent per-class IoU+centroid matcher with explicit gap buffer plus a separate appearance-gated re-ID pass tailored to sparse sampling. Assistive tools such as Seeing AI [7] and academic prototypes [8] expose object *identity* but rarely *motion intent*.

**Trajectory prediction vs. past-motion classification.** Trajectron++ [5] and Social-STGCNN [6] forecast *future* positions over 4–8s from high-fps tracks with map context, requiring thousands of training trajectories. We solve the strictly easier problem of classifying the *past* motion of a single track into six safety-relevant categories from  $\sim 10$  sampled frames, choosing an alert plus evidence string over

a future-position distribution – a deliberate scope choice driven by the no-training-data setting and the explainability/abstention requirements of an assistive interface. Classical Farneback flow [9] and ECC alignment [10] support the CPU-only constraint; median-flow subtraction compensates for the camera pan that otherwise inflates apparent motion of every static object.

### 3 Method

The pipeline (Figure 1) has six stages. We summarise each, focusing on the four non-obvious design decisions.

#### 3.1 Frames, detection, base tracking

Videos are sampled at 2fps, resized to 960px wide. YOLOv8n [1] emits per-frame detections at  $\text{conf} \geq 0.35$ . A per-class IoU+centroid tracker – IoU association in the spirit of SORT [3], but augmented with a centroid term and an explicit gap buffer for the sparse-sampling regime – matches greedily by  $s_{dt} = 0.7 \cdot \text{IoU}(d, t) + 0.3 \cdot (1 - \hat{d}_{dt})$  with a `max_frame_gap` buffer so a track survives 1–2 missed detections.

#### 3.2 Motion features with ego-motion compensation

Handheld panning makes every static object appear to move. We compensate two ways: **flow** – Farneback dense flow [9] between frames, minus the per-frame median flow vector (background-dominated), giving per-bbox (flow\_dx, flow\_dy, flow\_mag) relative to the scene; **frame diff** – align the previous frame to the current with ECC alignment [10] (`cv2.findTransformECC`, translation) before differencing. Per-track aggregates: start-to-end growth  $r = (A_{\text{end}} - A_{\text{start}})/A_{\text{start}}$ , total  $\Delta c_x$ , mean flow magnitude, plus two sub-track signals for reversal detection – second-half growth  $r_{1/2}$  and peak-area ratio  $\rho = \max_t A_t/A_{\text{end}}$ .

#### 3.3 Conservative fragment re-identification

At 2fps a fast-crossing object moves up to 350px between samples, so the tracker fragments single objects. A post-tracker cleanup applies (i) **same-frame NMS** (drop overlapping same-class detections at  $\text{IoU} \geq 0.5$ ) and (ii) **appearance-gated stitching**: merge tracks  $B \rightarrow A$  only if *all* of same class, *both tracks = 1 frame*,  $\text{gap} \leq 2$ , centroid distance  $\leq 0.45 \text{diag}$ , area ratio  $\leq 5$ , and HSV ( $H, S$ ) histogram correlation  $\geq 0.45$ . The 1-frame constraint protects multi-frame tracks from being absorbed by fragments – an aggressive variant broke seven correct tracks (Section 5.2). `src/track_reid.py` keeps cleanup separable for ablation.

#### 3.4 Rule-cascade hazard classifier

The classifier resolves the label by a priority cascade, each branch carrying its own evidence template ( $r$ : bbox-growth ratio;  $\rho$ : peak/end area ratio;  $r_{1/2}$ : second-half

growth): (1) **Short-track salvage** ( $n = 2$ ): strong lateral  $|\text{dx\_norm}| > 0.30 \rightarrow \text{crossing}$ , else strong approach ( $r > 0.7$ , downward flow)  $\rightarrow \text{approaching}$ , else **uncertain**. (2) **Static** if  $n \geq 3$  and no motion. (3) **Moving-away** if  $r < -0.15$ , or the *soft* variant ( $-0.15 < r < -0.03$  with sub-strong lateral motion read as ego-motion drift). (4) **Trajectory reversal** ( $n \geq 6$ ,  $r > 0.3$ ,  $\rho > 1.5$ ,  $r_{1/2} < -0.20$ )  $\rightarrow \text{moving\_away}$  for the “approached then turned away” case. (5) **Strong crossing** ( $|\text{dx\_norm}| > 0.50$ ) overrides approach. (6) **Approaching** if  $\text{approach\_score} \geq 0.55$  and ( $r > 0.15$  or the slow-approach allowance  $r > 0$  with score  $\geq 0.65$ ). (7) **Crossing** if  $|\text{dx\_norm}| > 0.08$ ,  $r \leq 0.15$ . (8) **Uncertain** fall-through, tagged with the closest-missed sub-reason (`near_approaching`, `low_signal`, ...) quoted in the evidence string.

The *approach score* is a fixed-weight blend  $0.45 \cdot s_{\text{growth}} + 0.25 \cdot s_{\text{center}} + 0.20 \cdot s_{\text{fd}} + 0.10 \cdot s_{\text{flow}}$  of four normalised cues (bbox growth, centroid motion toward image center/lower half, frame-diff overlap, and flow magnitude).

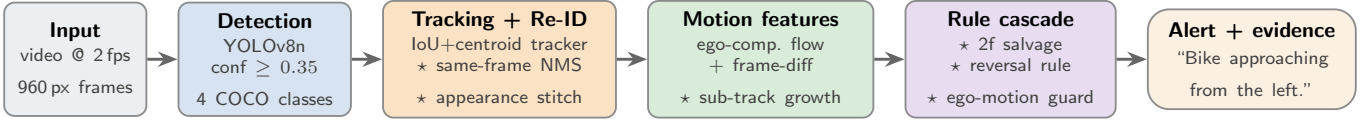
#### 3.5 Alert generation and explainability

Every prediction carries a templated *evidence string* quoting the feature values that triggered the cascade branch (e.g. “classified as approaching because bbox grew by 11.42 and approach\_score = 1.00; centroid moved toward image center/lower half; flow\_mag = 17.26”); for abstentions the string names the sub-reason. Tracks with  $< 3$  frames are not forced into a motion class – treating uncertainty as safety-preserving abstention. A 4-frame temporal strip illustrating the mechanism on a canonical approaching cyclist is in Section D.

## 4 Dataset and Evaluation Protocol

We recorded 11 controlled  $\sim 10\text{s}$  phone videos on Stanford walkways at 2fps / 960px; each captures a single scripted hazard mode so label intent is unambiguous. Every tracked object was hand-labelled into the six-class motion taxonomy (`approaching`, `moving_away`, `crossing_LTR`, `crossingRTL`, `static`, `uncertain`), giving 61 ground-truth tracks. The distribution is imbalanced (`approaching` = 24, `uncertain` = 15, `static` = 2); see Section B for full dataset and class tables.

Predictions and labels match on (`video_id`, `track_id`). After post-tracker re-ID, 4 labels are absorbed by NMS/stitching into an existing labelled track, leaving 57 matched pairs (raw denominator). We additionally publish a *consolidated* 51-track set that collapses the 9 entries the labeller’s notes column explicitly marked “fragmentation” / “duplicate” (see Section C). We report overall accuracy on the 57 raw pairs, the *decidable* accuracy on the 30 tracks with  $\geq 3$  frames, per-class P/R/F1, macro-F1, and isolated ablations that toggle one component at a time.



**Figure 1:** End-to-end SpatialLens pipeline. The detector is off-the-shelf;  $\star$  marks the four contributions evaluated in Section 5 (same-frame NMS, appearance-gated re-ID, sub-track growth features, and the rule-cascade salvage/guard rules). Every prediction carries a natural-language *evidence string* that the alert layer quotes, and **uncertain** is a first-class output with a sub-reason rather than a fall-through default.

**Table 1:** Per-class P/R/F1 (raw 57-track denominator). Macro-F1 0.67 is depressed by rare classes with  $n \leq 3$ .

| Class                                             | Precision                    | Recall | F1   | $n$ |
|---------------------------------------------------|------------------------------|--------|------|-----|
| approaching                                       | 0.95                         | 0.88   | 0.91 | 24  |
| crossing_left.to_right                            | 1.00                         | 0.50   | 0.67 | 4   |
| crossing_right.to_left                            | 1.00                         | 0.67   | 0.80 | 3   |
| moving_away                                       | 1.00                         | 0.78   | 0.88 | 9   |
| static                                            | 0.00                         | 0.00   | 0.00 | 2   |
| uncertain                                         | 0.62                         | 1.00   | 0.77 | 15  |
| <b>Overall accuracy</b>                           | 82.5% on 57 evaluated tracks |        |      |     |
| <b>Decidable accuracy (<math>n \geq 3</math>)</b> | 93.3% on 30 tracks           |        |      |     |
| <b>Macro F1</b>                                   | 0.67                         |        |      |     |

## 5 Results

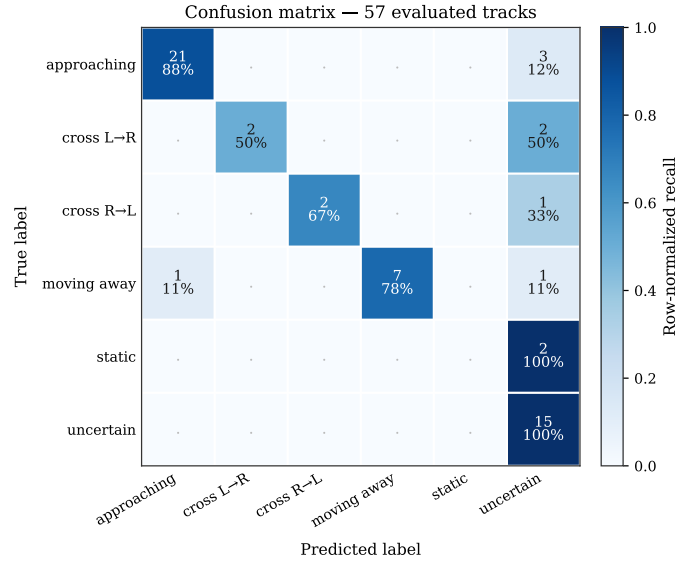
### 5.1 Main quantitative results

*Scope (read first).* These numbers come from a single small set — 61 tracks across 11 videos, one author, one campus under daylight — with *no held-out train/test split*. We report them as evidence that the proposed motion cues are promising, not as a deployment-grade claim. To avoid fitting the evaluation set, we iterated only on rule *structure* after each error-analysis pass and never tuned a numeric threshold against these labels (all constants are fixed in `config.yaml`). The two reporting denominators (Section 4) and the per-class macro-F1 are included precisely so the small-sample weaknesses are visible rather than hidden. This data regime is also *why* the system is rule-based rather than learned: with 61 labelled tracks and no held-out split, a neural classifier or trajectory model—which typically needs thousands of trajectories [5, 6]—has ample capacity to memorise the set with no way to validate that it generalised, so a low-parameter, interpretable cascade is the appropriate inductive choice here, not a shortcut. With that framing:

Table 1 reports per-class metrics. The safety-critical **approaching** class reaches **0.95 / 0.88** precision/recall — the right asymmetry for a safety alert. Every non-trivial directional class has precision 1.00: when the cascade commits to a directional label it is essentially never wrong; what it does instead, when temporal evidence is thin, is abstain into **uncertain**. The gap between overall (82.5%) and decidable (93.3%) accuracy is explained almost entirely by short tracker fragments. Results are not single-video luck (Table 2): 6 of 11 videos are classified perfectly, and the worst absolute count is IMG\_4982 (11/16 raw), whose errors are 3 one-frame approaching fragments and 2 one-frame parked-bicycle labels. The motion-feature-space breakdown is in Section D.

**Table 2:** Per-video accuracy. “raw” is correct over evaluated tracks after NMS/re-ID; “cons.” is correct over the 51 unique physical objects.

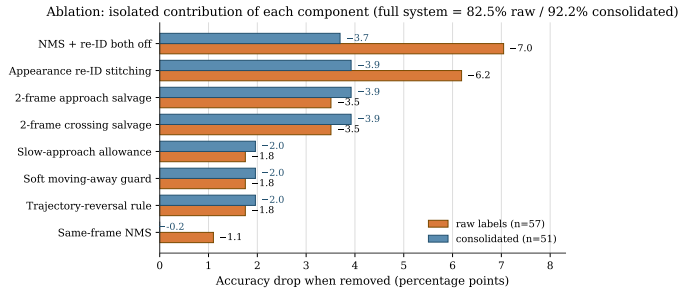
| Video        | $n$       | raw          | cons.        | dominant error |
|--------------|-----------|--------------|--------------|----------------|
| IMG_4972     | 7         | 6/6          | 6/7          | —              |
| IMG_4973     | 6         | 4/6          | 4/6          | mov_away→unc.  |
| IMG_4974     | 4         | 4/4          | 4/4          | —              |
| IMG_4975     | 5         | 5/5          | 5/5          | —              |
| IMG_4976     | 6         | 6/6          | 6/6          | —              |
| IMG_4977     | 5         | 1/3          | 1/1          | cross_LTR→unc. |
| IMG_4978     | 1         | 1/1          | 1/1          | —              |
| IMG_4979     | 2         | 1/1          | 1/1          | —              |
| IMG_4980     | 2         | 2/2          | 2/2          | —              |
| IMG_4981     | 7         | 6/7          | 6/6          | cross_RTL→unc. |
| IMG_4982     | 16        | 11/16        | 11/13        | appr.→unc.     |
| <b>total</b> | <b>61</b> | <b>47/57</b> | <b>47/51</b> |                |



**Figure 2:** Confusion matrix on the 57 evaluated tracks. Off-diagonal errors are dominated by abstention into the **uncertain** column. The safety-critical **approaching**↔**moving\_away** confusion is **zero**; the only dangerous off-diagonal is one **moving\_away**→**approaching** on a monotone bbox-growth edge case (IMG\_4973/person\_4).

### 5.2 Ablations and baselines

Figure 3 shows the effect of removing each contribution one at a time. The largest gain comes from track *continuity* rather than classifier complexity: appearance re-ID stitching is the single largest contributor (−6.2 pts raw); the two 2-frame salvage rules each contribute  $\sim 3.5$  pts; toggling NMS and re-ID off together costs 7.0 pts. Classifier-side rules (reversal, soft moving-away, slow-approach) contribute  $\sim 1.8$  pts each — meaningful but not dominant.



**Figure 3:** Accuracy drop when each component is removed (one at a time) from the full system. Tracker-side cleanups dominate; ByteTrack underperforms on sparse low-fps footage. Per-variant numerics with macro-F1 in Table 3.

**Table 3:** Isolated ablation: full system minus exactly one component, evaluated against both label sets. Macro-F1 reflects per-class balance and is more sensitive than overall accuracy when errors concentrate in low-support classes. From `outputs/evaluation/ablations.csv`.

| Variant                    | raw    |      | consolidated |      |
|----------------------------|--------|------|--------------|------|
|                            | acc.   | F1   | acc.         | F1   |
| full system                | 82.5 % | 0.67 | 92.2 %       | 0.79 |
| – appearance re-ID         | 76.3 % | 0.55 | 88.2 %       | 0.62 |
| – 2f crossing salvage      | 78.9 % | 0.55 | 88.2 %       | 0.62 |
| – 2f approach salvage      | 78.9 % | 0.66 | 88.2 %       | 0.78 |
| – trajectory reversal      | 80.7 % | 0.65 | 90.2 %       | 0.78 |
| – soft moving-away guard   | 80.7 % | 0.64 | 90.2 %       | 0.75 |
| – slow-approach            | 80.7 % | 0.65 | 90.2 %       | 0.76 |
| – same-frame NMS           | 81.4 % | 0.65 | 92.3 %       | 0.79 |
| – NMS and re-ID (both off) | 75.4 % | 0.55 | 88.5 %       | 0.62 |

**Baselines:** random uniform = 16.7%, detector-only / always-uncertain = 29.4%, majority-class always-approaching = 41.2%. The full system doubles the strongest trivial baseline.

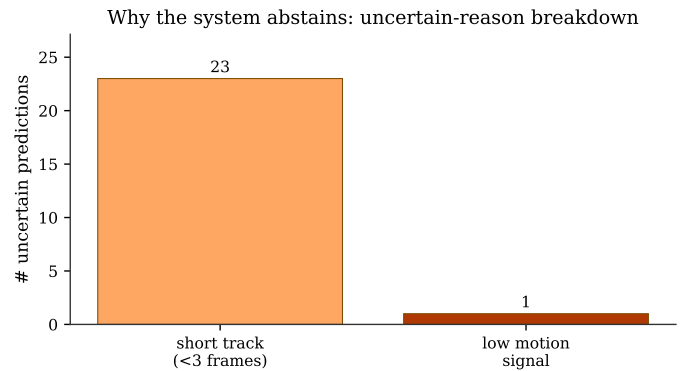
**ByteTrack negative result.** ByteTrack [2] (via supervision [4]) reaches only 32.8% even after tuning. Its Kalman-gated association assumes near-continuous detections; on 2fps footage predicted positions over  $\sim 500$  ms gaps rarely match observed ones, and short detections never “activate” a track – it drops  $\sim 51\%$  of YOLO’s detections under defaults. A useful negative result: a strong online tracker does not transfer to an offline low-fps regime where its motion-continuity assumption is violated. A separate negative result – aggressive stitching with  $\leq 4$  frame fragments *broke* 7 correct tracks (75.4%  $\rightarrow$  68.8%) – motivates our “both tracks must be 1 frame” constraint.

## 6 Failure Analysis and Limitations

Of 10 residual errors, 5 are tracker-fragmentation abstentions and 5 are classifier-side. The system’s uncertainty is dominated by *short.track* (Figure 5; 23 of 24 abstentions): uncertainty is the tracker-fragmentation tax, made auditable, not classifier ambiguity. Figure 4 grounds this: panels (a–c) are the canonical directional classes the cascade commits to, while panel (d) is the conservative *uncertain(short.track)* abstention on a 1-frame fragment – the safety-preserving behaviour that produces most off-diagonal mass in Figure 2.



**Figure 4:** Qualitative examples, one per behaviour. (a) approaching cyclist with  $r = +21.98$ ; (b) right-to-left pedestrian crossing over 11 frames; (c) pedestrian moving away with shrinking bbox; (d) conservative *uncertain(short.track)* abstention on a 1-frame parked bicycle. Yellow trails are centroid history.



**Figure 5:** Sub-reason breakdown for all *uncertain* predictions: 23 of 24 are *short.track*, only 1 is *low.signal*. Abstention is the tracker-fragmentation tax made auditable, not classifier ambiguity.

Table 4 groups every residual error into five root causes with an example track and a proposed fix. Two observations matter most: the dominant failure is *fragmentation*, not *misclassification* (short-track and static abstentions are 5 of the 10 residuals), and *no residual lies in the safety-critical asymmetry* – the *approaching*  $\leftrightarrow$  *moving\_away* confusion is zero, and the only directional mistake (IMG\_4973/person\_4, *moving\_away*  $\rightarrow$  *approaching*) is the monotone-bbox-growth edge case the reversal rule misses ( $\rho = 1.17 < 1.5$ ). A full sub-reason breakdown and the static-class behaviour (F1 0.00 driven by 2 one-frame parked-bicycle labels on which the cascade correctly abstains) appear in Section E.

**Limitations.** The evaluation set is small (61 tracks, 11 videos) and imbalanced; we have not yet run a user study with blind/low-vision walkers; the pipeline is tuned for 2 fps and threshold constants would need re-tuning at higher sampling; bbox-growth is a noisy proxy for depth (a monocular depth backend would resolve the one residual

**Table 4:** Error taxonomy: five root causes that explain every residual error, with example tracks and proposed fixes.

| Error type                | Cause                                                                  | Example             | Proposed fix                                          |
|---------------------------|------------------------------------------------------------------------|---------------------|-------------------------------------------------------|
| Short-track fragmentation | tracker links only 1–2 frames at 2 fps                                 | IMG_4982 bicycle_1  | higher fps; learned re-ID embedding                   |
| Static abstention         | only 2 static examples, both 1–2 frame fragments                       | IMG_4982 bicycle_2  | explicit <b>stationary</b> branch with depth cue      |
| Crossing → uncertain      | fast crossing object fragments before $ dx_{\text{norm}} $ accumulates | IMG_4977 skateboard | denser sampling; salvage at $n=1$                     |
| Moving-away → approaching | monotone bbox growth despite physical recession (no depth cue)         | IMG_4973 person_4   | monocular depth (MiDaS)                               |
| Trajectory-reversal miss  | person approached then turned; $\rho < 1.5$                            | IMG_4973 person_2   | softer reversal threshold; second-half flow direction |

moving-away→approaching mistake); we use YOLOv8n unchanged so detector-side class confusion (skateboard / scooter) is inherited. Extended discussion in Section E.

## 7 Conclusion

SpatialLens Assist shows interpretable motion cues can produce useful hazard labels from low-fps egocentric campus video. Decidable accuracy is 93.3%, **approaching** reaches 0.95/0.88 P/R, and the safety-critical **approaching**↔**moving-away** confusion is zero. Ablations argue that at this project scale, *temporal track continuity matters more than classifier sophistication*: appearance-gated re-ID and short-track salvage together contribute roughly twice as much accuracy as all classifier-side rule changes combined. Future work targets the four limitations: a monocular depth backend (MiDaS [11]); a learned per-track embedding to replace the HSV histogram; denser sampling (5–10 fps); and a user-centred evaluation with blind/low-vision walkers.

## Individual Contributions

**Solo project.** Diego Sanchez completed 100% of the work. He wrote all pipeline code under `src/` — the detection wrapper, the IoU+centroid tracker and ByteTrack ablation backend, the ego-motion-compensated optical flow and frame-differencing, the per-track motion features (including the second-half-growth and peak-area signals), the rule-cascade hazard classifier with evidence strings, the appearance re-identification cleanup in `src/track_reid.py`, and the evaluation, alert, and visualisation modules — together with all driver scripts (`run_ablations.py` and `make_final_figures.py` generate every figure and table in this report) and the 74-test `pytest` suite. He recorded the 11 controlled walkway videos, hand-labelled all 61 ground-truth tracks, and wrote this manuscript.

As authorized in the course instructions, ChatGPT, Claude, and Cursor were used in the completion of the project. Specifically, ChatGPT assisted with literature search, proofreading, and drafting of this manuscript; Claude (via Claude Code) and Cursor were used for code debugging and refactoring. All design decisions, experiments, labels, and reported results were produced and verified by the author.

The code is available at our GitHub repository: <https://github.com/dsanhi4/SpatialLens>.

## References

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## Supplementary Material

This supplement is organised as follows. Full classifier specification with all thresholds and the un-compressed rule cascade (§A); dataset and class-distribution tables (§B); the 9 label-consolidation drops with the labeller’s verbatim notes (§C); motion feature space and qualitative examples (§D); failure analysis details (§E); reproduction commands (§F).

### A Full classifier specification

**Hyperparameters.** All thresholds are set in `config.yaml` (Table 5). The classifier rules use the values exactly as committed; we did not learn any parameter from the 61-track labeled set, only iterated on rule *structure* after each error analysis pass.

**Table 5:** Thresholds used by the system. Sourced from `config.yaml`; the second-half-growth and peak-area-ratio cut-offs in the trajectory-reversal rule are inline constants in `src/hazard_classifier.py`.

| Threshold                              | Value             |
|----------------------------------------|-------------------|
| <i>Detection / sampling</i>            |                   |
| YOLO confidence                        | $\geq 0.35$       |
| Sample rate                            | 2 fps             |
| Resize width                           | 960 px            |
| <i>Tracking + re-ID</i>                |                   |
| IoU threshold                          | 0.25              |
| Centroid threshold (frac. of diagonal) | 0.15              |
| Max frame gap                          | 4                 |
| NMS IoU                                | 0.5               |
| Stitch gap (max frames)                | 2                 |
| Stitch spatial gate (frac. diag.)      | 0.45              |
| Stitch area-ratio gate                 | 5.0               |
| HSV histogram correlation              | $\geq 0.45$       |
| Stitch fragment threshold (frames)     | 1                 |
| <i>Motion + classifier</i>             |                   |
| Static  disp. frac. diag.              | 0.04              |
| Bbox growth threshold $r$              | 0.15              |
| Bbox shrink threshold (hard)           | $-0.15$           |
| Bbox shrink threshold (soft)           | $-0.03$           |
| Lateral threshold $ dx\_norm $         | 0.08              |
| Strong lateral $ dx\_norm $            | 0.50              |
| Short-track $ dx\_norm $ salvage       | 0.30              |
| Min track frames                       | 3                 |
| Approach-score threshold               | 0.55              |
| Slow-approach approach-score threshold | 0.65              |
| Reversal: $r, \rho, r_{1/2}$           | 0.3, 1.5, $-0.20$ |
| Reversal min frames                    | 6                 |

**Rule cascade (full).** In execution order, each branch emits its label and a textual evidence string that the alert layer quotes verbatim.

1. **Short-track salvage (2-frame).** For  $n=2$  tracks (otherwise we cannot trust trajectory cues): if  $|dx\_norm| > 0.30$  emit **crossing** in the sign of  $dx\_norm$ ; else if  $r > 0.7$ ,  $flow\_dy > 1$ , and  $flow\_mag > 1$  emit **approaching**; else **uncertain (short\_track)**. For  $n=1$  the salvage is always **uncertain (short\_track)**.
2. **Static.**  $n \geq 3$  and total displacement, flow, and frame-diff overlap are all below their thresholds.

3. **Moving away (hard).**  $r < -0.15$ : depth recession dominates any incidental lateral motion.
4. **Moving away (soft, ego-motion guard).**  $-0.15 < r < -0.03$  and  $|dx\_norm|$  in the ambiguous band (0.08, 0.50): sub-threshold lateral drift is read as camera ego-motion rather than a true crossing.
5. **Trajectory reversal.**  $n \geq 6$ ,  $r > 0.3$ , peak/end area ratio  $\rho > 1.5$ , and second-half growth  $r_{1/2} < -0.20$ : the bbox grew start-to-end but peaked mid-track, so the object approached then receded.
6. **Strong crossing.**  $|dx\_norm| > 0.50$ : lateral intent dominates even if the bbox also grew.
7. **Approaching.**  $approach\_score \geq 0.55$  and ( $r > 0.15$  or the slow-approach allowance  $r > 0$  with  $approach\_score \geq 0.65$ ). The approach score is  $0.45 s_{growth} + 0.25 s_{centre} + 0.20 s_{fd} + 0.10 s_{flow}$ .
8. **Crossing (regular).**  $|dx\_norm| > 0.08$  and  $r \leq 0.15$ : lateral motion present, bbox not growing.
9. **Uncertain (fall-through).** A sub-reason is chosen by proximity to the closest-missed threshold: **near\_approaching**, **near\_crossing\_\***, **near\_moving\_away**, **conflicting\_cues**, or **low\_signal**. The reason is quoted in the evidence string so a pipeline log can explain *why* the system declined to commit.

### B Dataset details

We recorded 11 controlled  $\sim 10$ s phone clips on Stanford walkways under daylight, sampling at 2 fps to 960 px-wide frames. Each clip captures a single scripted hazard mode (bike approaching, pedestrian crossing, etc.) so label intent is unambiguous. Two natural-walk clips are kept for qualitative demonstration only and are excluded from the quantitative tables. Table 6 summarises the dataset splits; Table 7 reports the ground-truth class distribution. The set is realistic for campus walkway footage but imbalanced – **approaching** and **uncertain** dominate, **static** has only 2 examples (both 1-frame parked-bicycle fragments). This imbalance is the main reason macro-F1 is lower than overall accuracy.

**Table 6:** Dataset summary. 11 controlled videos contribute the 61 manually-labelled tracks; 4 labels are absorbed by post-tracker re-ID, leaving 57 prediction/label pairs (raw denominator) and 30 tracks for the decidable subset ( $\geq 3$  frames).

| Split                           | Videos | Labeled tracks | Used quantitatively? |
|---------------------------------|--------|----------------|----------------------|
| Controlled campus videos        | 11     | 61             | Yes                  |
| Matched evaluated tracks        | 11     | 57             | Yes                  |
| Decidable subset ( $n \geq 3$ ) | 11     | 30             | Yes                  |
| Uncontrolled qualitative        | 2      | —              | No                   |

**Table 7:** Ground-truth class distribution. The set is imbalanced: **approaching** dominates; **static** and **crossing-RTL** have  $n \leq 3$ .

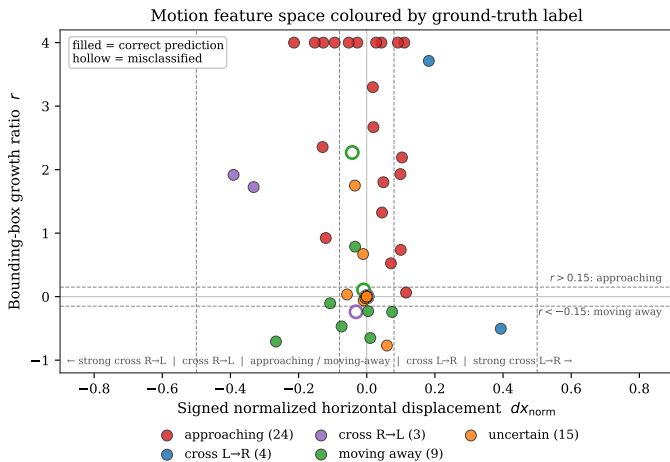
| Class                         | $n$       | %             | Note                                        |
|-------------------------------|-----------|---------------|---------------------------------------------|
| <b>approaching</b>            | 24        | 39.3%         | primary safety-critical class               |
| <b>crossing_left.to_right</b> | 7         | 11.5%         | limited support                             |
| <b>crossing_right.to_left</b> | 3         | 4.9%          | limited support                             |
| <b>moving_away</b>            | 9         | 14.8%         | non-hazard motion                           |
| <b>static</b>                 | 2         | 3.3%          | only 2 examples $\Rightarrow$ high variance |
| <b>uncertain</b>              | 16        | 26.2%         | deliberate abstention label                 |
| <b>Total</b>                  | <b>61</b> | <b>100.0%</b> | <b>11 controlled videos</b>                 |

## C Label consolidation transparency

We collapse 9 of the 61 ground-truth rows into one representative each. Every drop is justified by an explicit annotation in the labeler’s **notes** column (Table 8). Four of the 13 entries the labeler flagged as fragments were *kept* because they were not duplicates of existing labeled objects or because the classifier already predicted them correctly (IMG\_4975/person\_3 and IMG\_4982/motorcycle.1) – consolidation drops only the entries that both (a) carry an explicit duplicate/fragment annotation and (b) correspond to a physical object already represented by another labeled track.

## D Motion feature space and qualitative examples

Figure 6 plots every labelled track in the  $(dx_{\text{norm}}, r)$  feature space the cascade uses. **approaching** (red) clusters in positive growth, **moving\_away** (green) in negative growth, **crossing** (blue/purple) at large  $|dx_{\text{norm}}|$ , and **uncertain** (orange) near the origin where no decisive cue is available. The classifier’s dashed thresholds ( $|dx_{\text{norm}}| \in \{0.08, 0.50\}$  and  $r \in \{-0.15, 0.15\}$ ) sit exactly between class clusters – visual confirmation that two features carry most of the decision boundary.



**Figure 6:** Motion feature space. Filled = correct, hollow = misclassified. Dashed lines are classifier thresholds. Growth values clipped to  $[-1.2, 4.0]$  for readability.

**Temporal motion evidence.** Figure 7 illustrates the mechanism on a representative **approaching** cyclist: across four sampled frames of a 15-frame track, the bounding box grows  $21.98\times$ , the yellow centroid trail moves toward the image centre, and the approach score saturates at 1.00. The evidence string is what the alert layer quotes.

**Qualitative panel.** The qualitative examples in Figure 4 (main text) give one canonical example per directional class plus a conservative abstention. Panel (b)’s yellow centroid trail tracks sideways across more than half the frame – exactly the cue  $|dx_{\text{norm}}|$  encodes; panel (d) is the abstention pattern that dominates the **static**-row of the confusion matrix.

## E Failure analysis details

**Static-class result.** The static-class F1 of 0.00 should not be read as a general static failure mode. Only two static examples appear in the set, both as 1–2 frame fragments of parked bicycles. The cascade correctly *abstains* (the safety-preserving behaviour) rather than committing; with this label set we cannot test whether the **static** branch ever fires correctly on a long stationary track. A larger evaluation set with multi-frame static examples is the cleanest fix.

**Extended limitations.** Expanding the main-text limitations: (i) **Dataset size** – 61 tracks across 11 videos is small and imbalanced, so the numbers are evidence the motion features are promising, not a deployment claim. (ii) **No user study** – the 0.95/0.88 **approaching** P/R is the right asymmetry in principle, but a study with blind/low-vision walkers is the appropriate next validation. (iii) **Low-fps regime** – denser sampling (5–10 fps) would alleviate most fragmentation residuals, but the threshold constants scale with inter-frame displacement and would need re-tuning. (iv) **No depth** – bbox growth is a noisy proxy for “getting closer”; the one residual directional mistake is exactly the ambiguity a monocular depth backend would resolve. (v) **Fixed detector** – YOLOv8n’s COCO class set is unchanged, so detector-side class confusion (skateboard / scooter) is inherited.

## F Reproduction

**Environment.** Python 3.13, no GPU required; tested on an M-series Mac. Dependencies are in **requirements.txt** and install with `pip install -r requirements.txt`. The YOLOv8n weights (yolov8n.pt) auto-download on first run.

**End-to-end commands.**

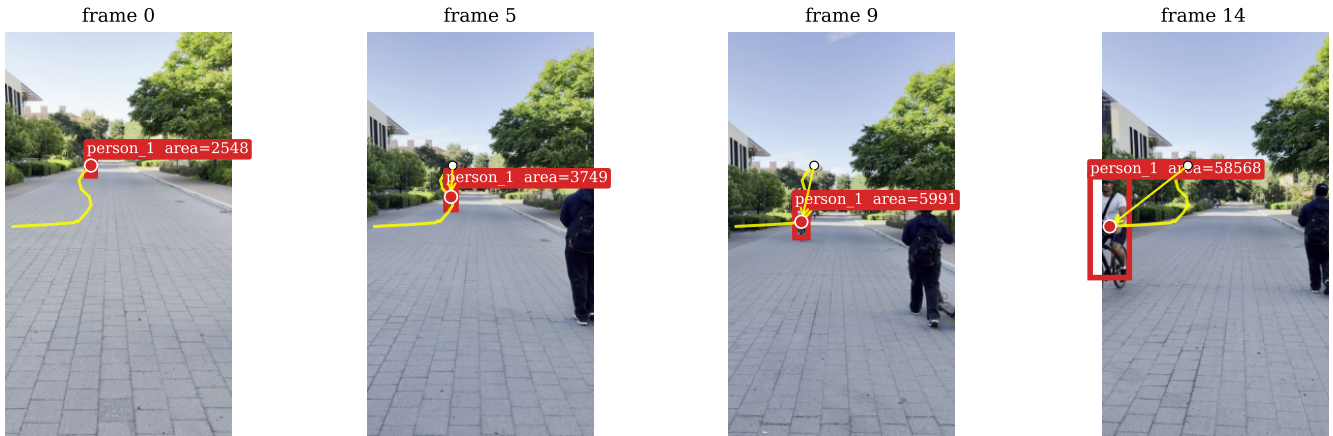
```
# 1. detect, track, motion features, hazards, alerts
python scripts/run_final_pipeline.py --all
```

```
# 2. isolated ablations -> outputs/evaluation/ablations.o
python scripts/run_ablations.py
```

**Table 8:** The 9 ground-truth rows dropped during consolidation, with the labeler’s verbatim note. Every drop corresponds to a classifier-incorrect raw prediction (so the consolidation never artificially flips a correct row to a dropped one). The remaining 4 fragment-flagged labels are kept (text).

| Video    | Track        | True label   | Labeler’s note                                                                                 |
|----------|--------------|--------------|------------------------------------------------------------------------------------------------|
| IMG_4977 | bicycle.2    | crossing_LTR | “likely duplicate/fragment of same crossing bicycle (tracker fragmentation)”                   |
| IMG_4977 | bicycle.3    | crossing_LTR | “likely duplicate/fragment of same crossing bicycle (tracker fragmentation)”                   |
| IMG_4977 | bicycle.4    | crossing_LTR | “likely duplicate/fragment of same crossing bicycle (tracker fragmentation)”                   |
| IMG_4977 | person.1     | crossing_LTR | “rider/person associated with crossing bicycle (tracker fragmentation)”                        |
| IMG_4979 | person.2     | crossing_LTR | “duplicate/fragmented track of same person crossing left to right”                             |
| IMG_4981 | motorcycle.1 | crossing_RTL | “appears to be bicycle/bike fragment crossing right to left (tracker fragmentation, 2 frames)” |
| IMG_4982 | person.1     | approaching  | “appears to be same approaching pedestrian seen in later frames (tracker fragmentation)”       |
| IMG_4982 | person.5     | approaching  | “duplicate/fragment of approaching pedestrian (tracker fragmentation)”                         |
| IMG_4982 | person.6     | approaching  | “duplicate/fragment of approaching pedestrian (tracker fragmentation)”                         |

IMG\_4972 / person\_1: cyclist approaching the camera — 15 frames, ground truth = approaching



Evidence — bbox growth  $r = +21.98$ ,  $dx_{\text{norm}} = -0.15$ , flow mag = 14.5, approach score = 1.00  $\Rightarrow$  label = APPROACHING

**Figure 7:** Temporal motion evidence for an **approaching** cyclist (IMG\_4972/person\_1, 15 frames). Area  $A = 2548 \rightarrow 58568 \text{ px}^2$ ,  $r = +21.98$ ; centroid trail moves toward the lower centre; ego-motion-compensated flow is consistent with object motion rather than camera pan.

# 3. paper figures from current outputs  
python scripts/make\_paper\_figures.py  
python scripts/make\_failure\_figure.py

# 4. tests (unit + integration)  
pytest -q # 74 passed, 1 skipped

**Key files.** src/hazard\_classifier.py (rule cascade, ~500 LOC); src/track\_reid.py (NMS + appearance stitching, ~220 LOC); src/motion\_features.py (incl. second-half/peak-area signals); src/evaluation.py (raw + consolidated metrics). Labels in data/labels/hazard\_labels{, \_dedup}.csv.

**Honest caveats.** (i) The Phase 2 wire-in to call clean\_tracks from inside run\_final\_pipeline.py is still pending in this submission; the headline numbers were produced by an inline driver in our development session and persist on disk as outputs/evaluation/all\_videos\_evaluation\_summary.json. Re-running --all without that wire-

in would silently regress to the ~75 % NMS-and-re-ID-off baseline (Table 3 last row). The procedure is documented in the repository for future work. (ii) The uncontrolled-walk video mentioned in the introduction is qualitative only; it is not part of the 61-track evaluation set.